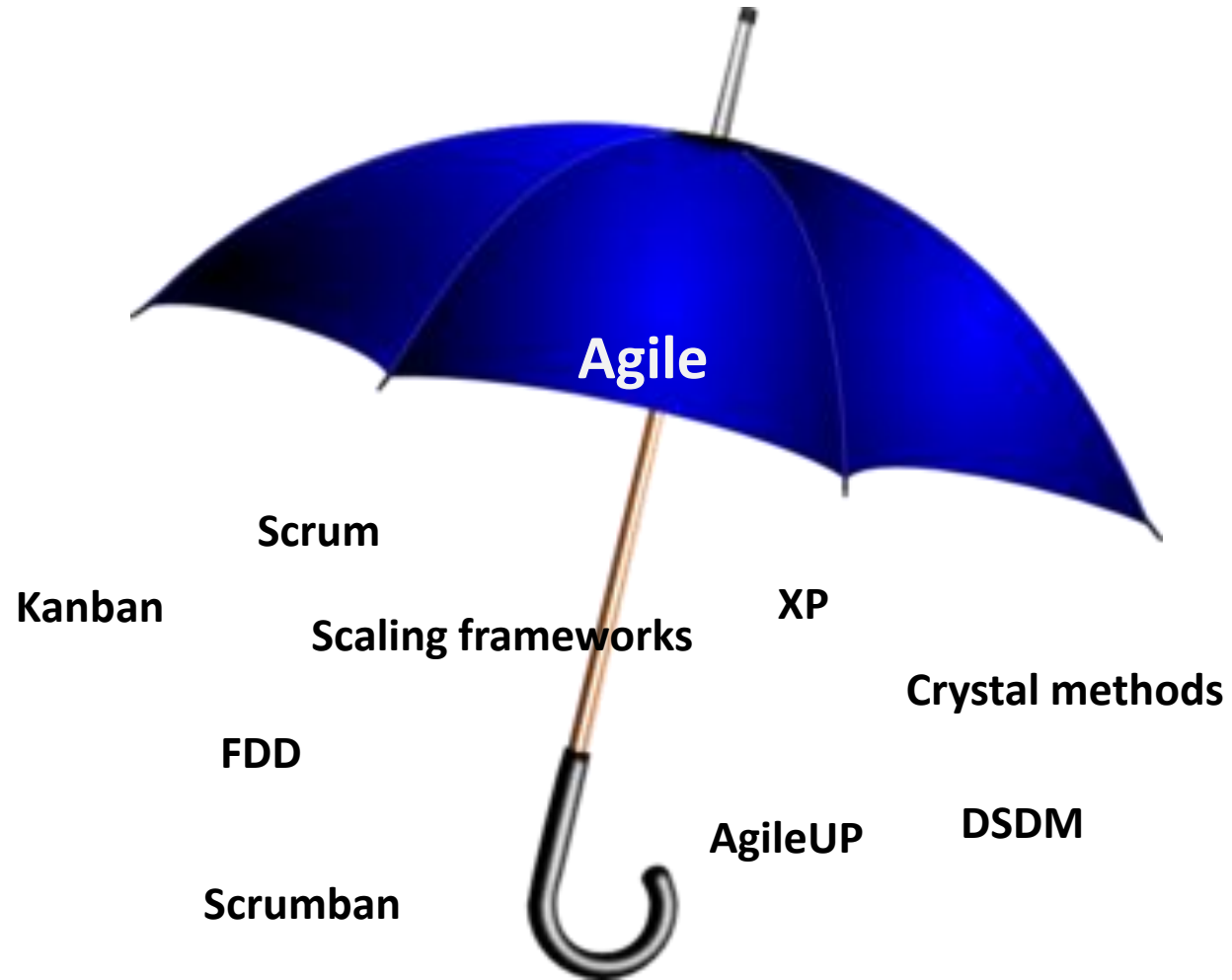


Agile and Requirement Engineering in Agile

Week 3

Dr. Natasha Ali Mian

Agile Umbrella



Fitting Agile Everywhere

ONE SIZE
DOESN'T FIT ALL



Agile methodology not suitable

- High certainty projects
- Services project
- Fixed price projects

Agile methodology suitable

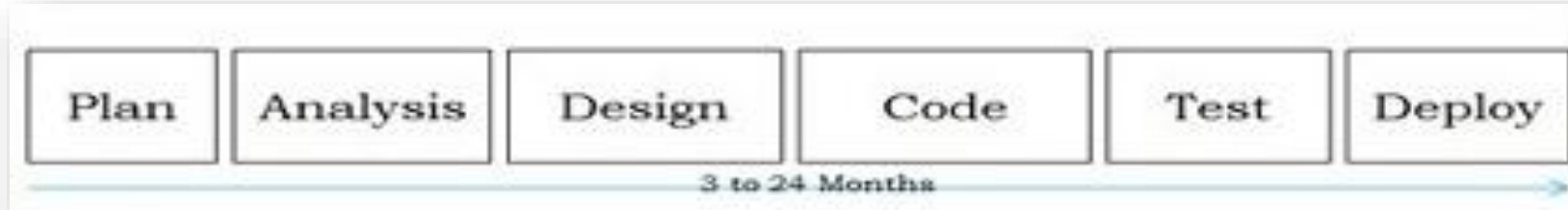
- Low certainty projects
- There is starting point but no predetermined scope
- Allow discovery for future evolution
- New startups

Project life cycles

- **Four types of life cycles**
 - **Predictive life cycle**
 - **Iterative life cycle**
 - **Incremental life cycle**
 - **Agile life cycle**

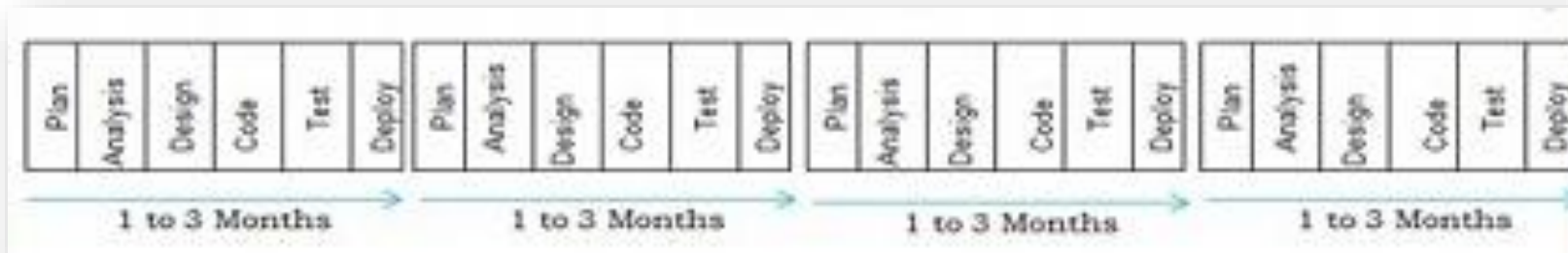
Predictive life cycle/Plan driven/waterfall

- A more traditional approach, with the bulk of planning occurring upfront, then executing in a single pass; sequential process
- Take advantage of known and proven things.
- Reduced uncertainty and complexity
- Allow team to segment work into a sequence of predictable grouping



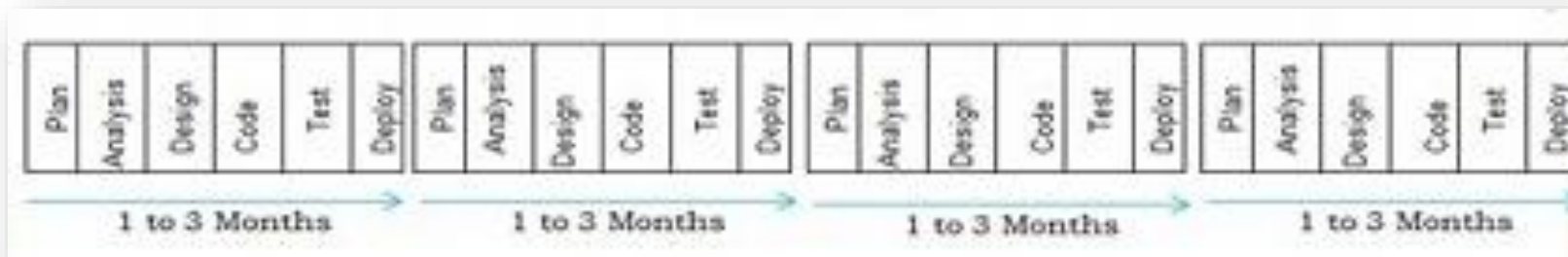
Iterative life cycle

- Allows feedback for unfinished work to improve and modify that work.
- Projects take advantage of iterative life cycle when complexity is high
- Used when changes are frequent.
- May take longer because optimized for learning rather than delivery



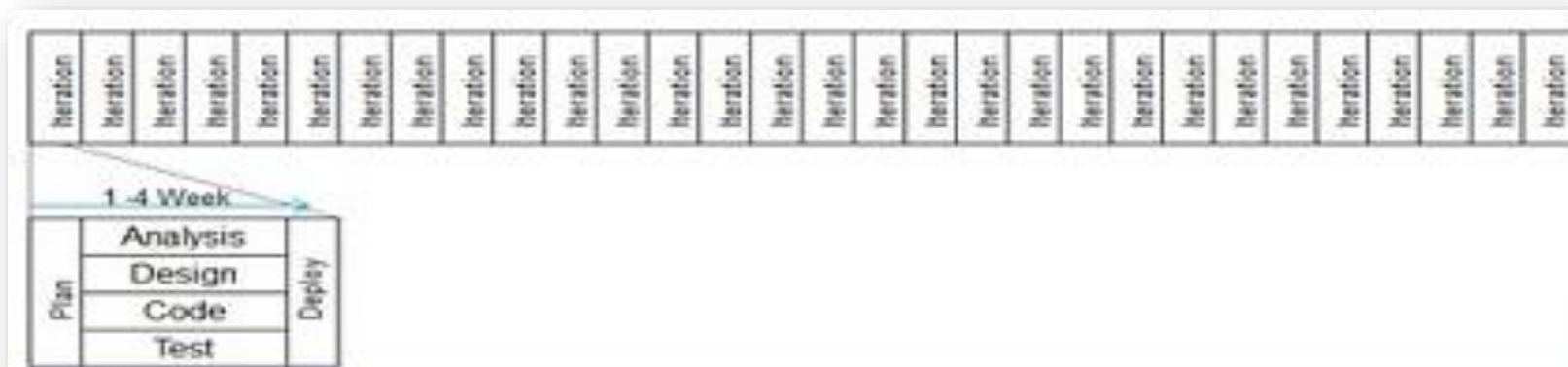
Incremental life cycle

- Frequent delivery of smaller deliverables is called an incremental life cycle
- An approach that provides finished deliverables that the customer may be able to use immediately.
- Customers are willing to take subset of overall solution
- Projects optimized for speed of delivery
- Businesses that couldn't wait for every thing to be completed



Agile life cycle/Change driven/Adaptive

- An approach that is
 - both iterative and incremental to refine work items ,
 - deliver frequently and
 - get feedback to better plan the next part of the project.
- Customer satisfaction increases with early and continues delivery of valuable product.
- In iteration based agile teams works in iteration(time boxes of equal duration) to deliver completed features.

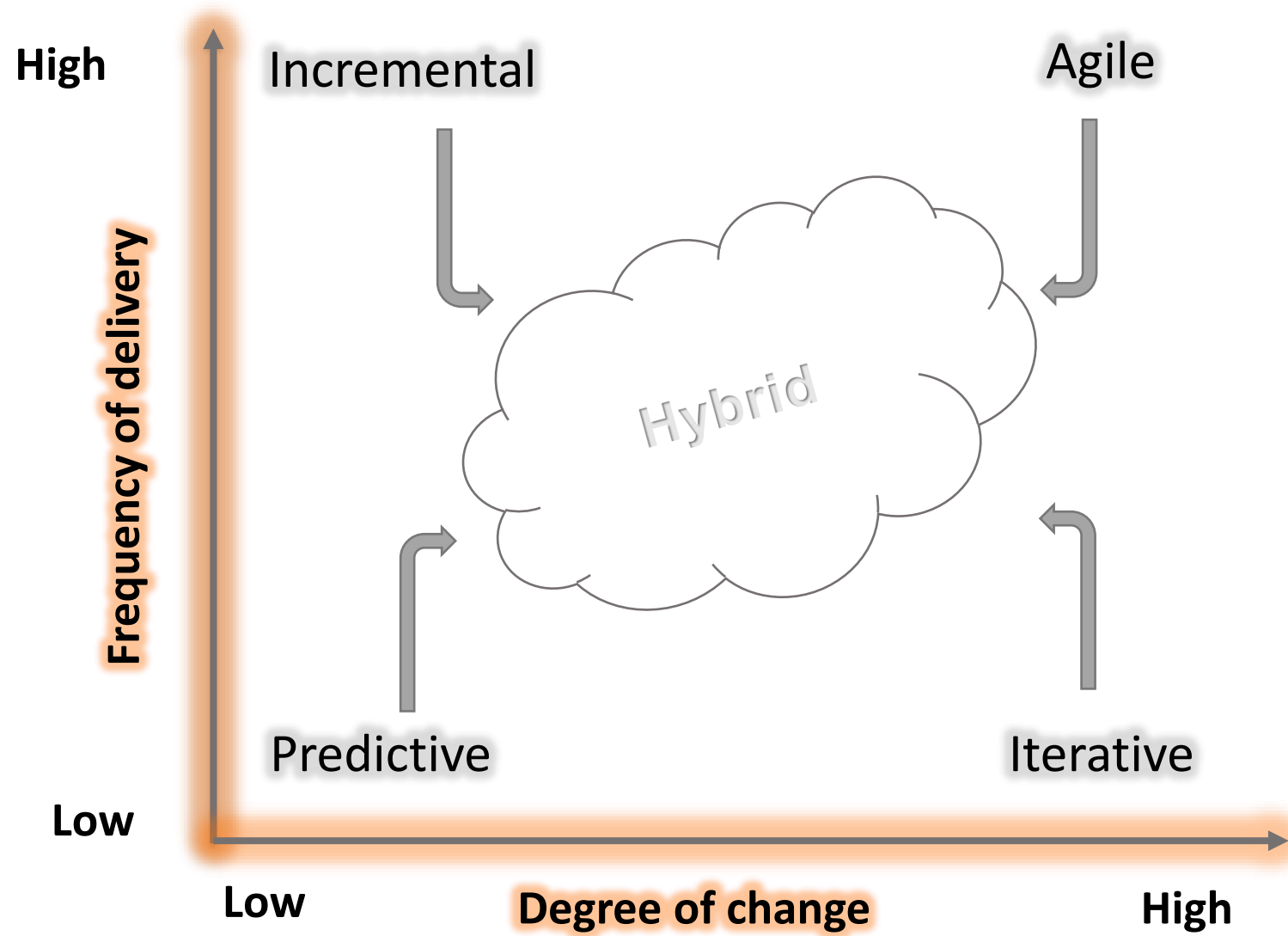


Characteristics of project life cycles

Life Cycle	Requirements	Activities	Delivery	Goal
Predictive	Fixed	Each activity performed one time for entire project	Single delivery	Manage goal and cost
Iterative	Changing	Activities repeated until things get correct	Single delivery	Correctness of solution
Incremental	Changing	Each activity performed one time for a given increment	Frequent smaller deliveries	Frequency/speed
Agile	Changing	Activities repeated until things get correct	Frequent smaller deliveries	Customer value via frequent deliverables and their feedback

Hybrid life cycle

- A combination of predictive, iterative, incremental and/or agile approaches is a hybrid approach.
- The point is to select a life cycle or a combination of life cycles that work for the project, the risk/team and the culture.
- Doing agile technique should not be the goal but to achieve project goals so its not necessary to use single approach for entire project.
- For example, a detailed requirements effort, followed by sprints of incremental delivery would be a “Hybrid Approach”.



Agile manifesto

Manifesto for Agile Software Development

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and interactions over **processes and tools**
Working software over **comprehensive documentation**
Customer collaboration over **contract negotiation**
Responding to change over **following a plan**

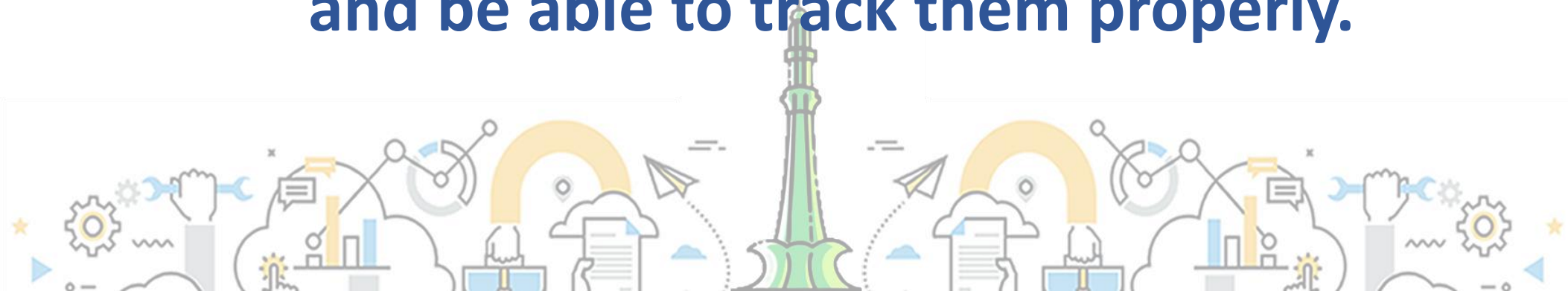
That is, while there is value in the items on the right, we value the items on the left more.



Requirement Pre-Requisites

What we hope to Achieve

**We will try to shed some light on
requirement engineering issues that would
help you in getting the right requirements
and be able to track them properly.**





Requirement Pre-Requisites

Before We Start!

A Stitch in Time Saves Nine!



Traditional Requirements

- Criteria to which system or business should adhere
- Generated before coding begins
- Often considered as constraints, conditions or capabilities of system
- Rarely contain acceptance criteria
- Contract between business and development

User Story

- A product feature description written by Customer/Proxy from the perspective of end user.
- It helps team understand what work is to be done
- Customer/Proxy prioritizes and team *estimates* them
- It usually follows the format:
 - As a : <Role>
 - I want to : <Function>
 - So that : <Desired Goal>

USER STORY TEMPLATE

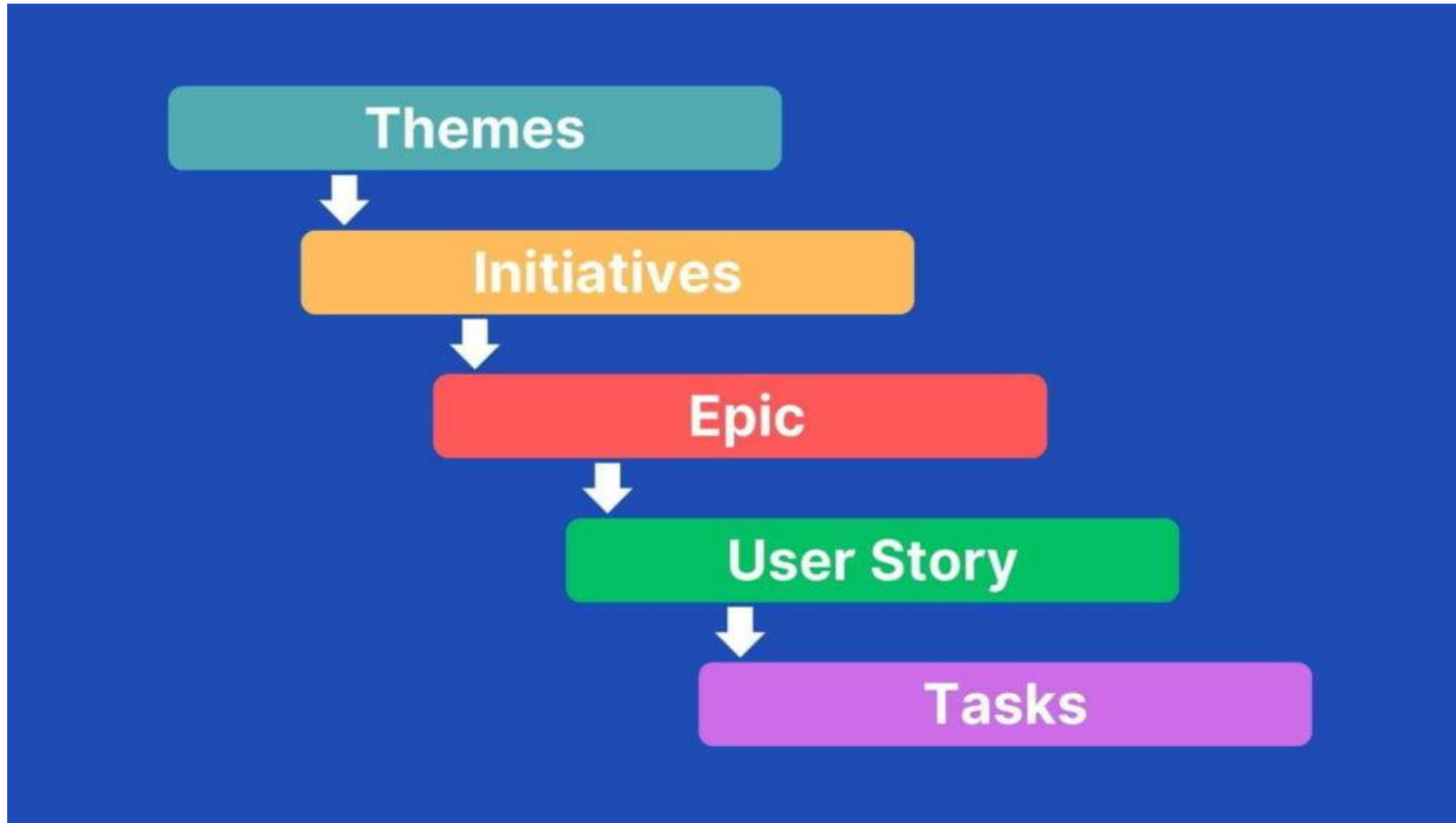
Story Title	User ID
User Story 1 As a(stakeholder) I want to (task), So That (desired result)	Importance Estimate
Acceptance Criteria Measurable results, what defines "done"? And I know I am done when	Type ☐ Payment ☐ Report/view ☐ Search ☐ Manage data ☐ Workflow 

User Story (contd.)

- Shifts focus from writing about requirements to talking about them
- Not a highly detailed document, rather a reminder for further collaboration
- A User Story should be End to End

INVEST

Independent	Reduced dependencies, easy planning
Negotiable	Details are added via collaboration
Valuable	Provides value to customer
Estimable	Should not be big / vague
Small	Can be done in less than a week by team
Testable	Has acceptance criteria





Epic and Theme

- **Epic** is a large story that can be further divided into multiple stories
- **Theme** is a collection of related stories
- For example, “A user can search for job” can be split into multiple stories:
 - Search by Location, Salary Range, etc.
 - View Job Detail
 - View Company Profile

Improve Student Experience -> Develop Student Portal-> View grades

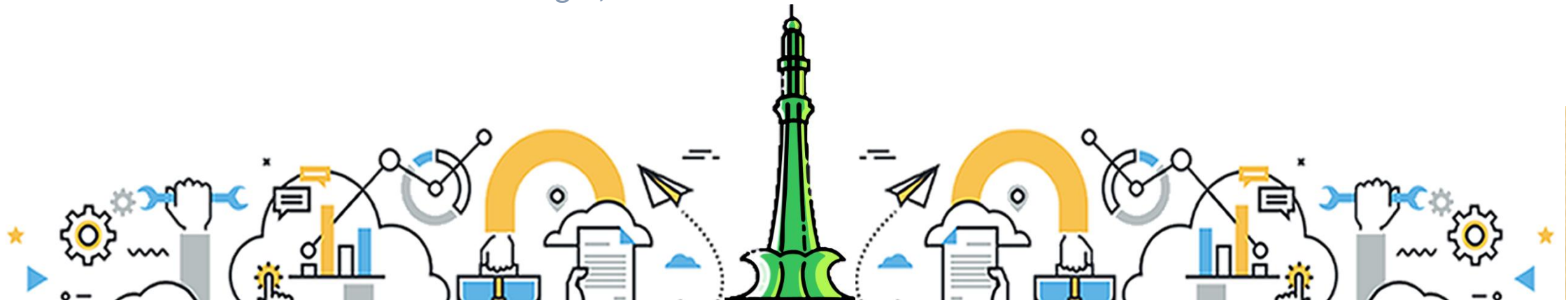


Requirement Pre-Requisites

Recap Agile Manifesto

- Individuals and **interactions** over processes and tools
- Working software over comprehensive documentation
- Customer **collaboration** over contract negotiation
- **Responding** to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.





Requirement Pre-Requisites

Requirement Sources

Although there is usually one main source but there are different types of that sources

1. Project Owner with Domain and Technical Knowledge
2. Project Owner with only Domain Knowledge
3. Project Owner with only Technical Knowledge
4. Project Owner – just because...





Requirement Pre-Requisites

Requirement Sources

Regardless of the source the main issue always come depending on
Location.

1. Local / Desi
2. English Speaking
3. Non-English Speaking (Who try very hard to speak English)



The Most Common Issue

What was required is not what was developed!





Requirement Pre-Requisites

WHY?!!!

1. Objective/Envisioned Product/Project was not clear
2. Requirements were not captured properly
3. Requirements understanding was different
4. Requirements were not QUESTIONED! Only questions about the requirements were asked.





Requirement Pre-Requisites

Solution 1/4

1. As we usually face communication hurdles we need to make sure that we Document the requirements given. This will ensure that all parties are on the same page

BUT THE MANIFESTO SAID Customer **collaboration** over contract negotiation
Or Working software over comprehensive documentation

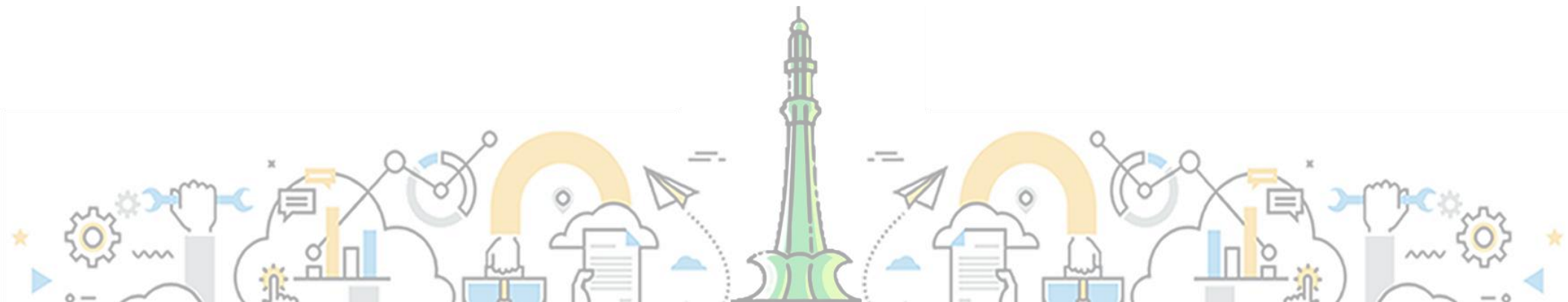




Requirement Pre-Requisites

Solution 2/4

2. Question the Requirements for deeper understanding of the Domain. Contribute in the Solution as it will help in covering scenarios that the project owner might have missed.

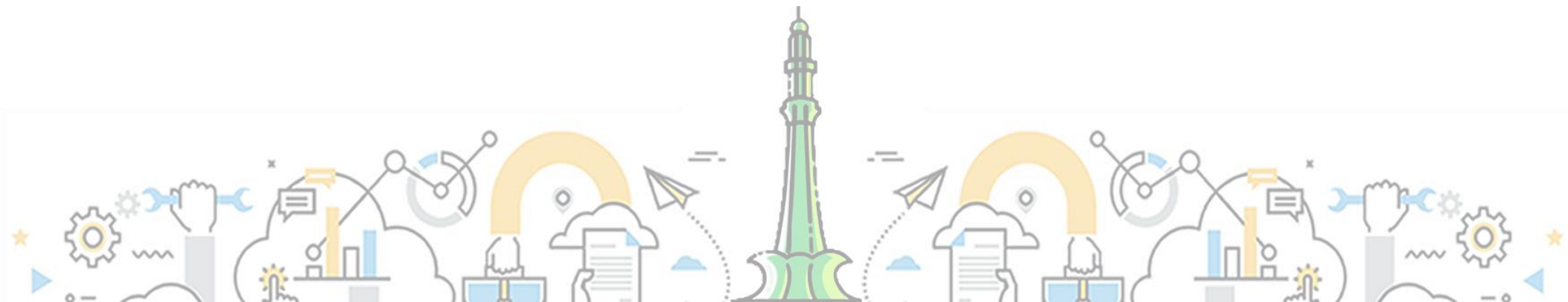




Requirement Pre-Requisites

Solution 3/4

3. Don't just jump into it. Lincoln Said "Give me six hours to chop down a tree and I will spend the first four sharpening the axe." Plan.

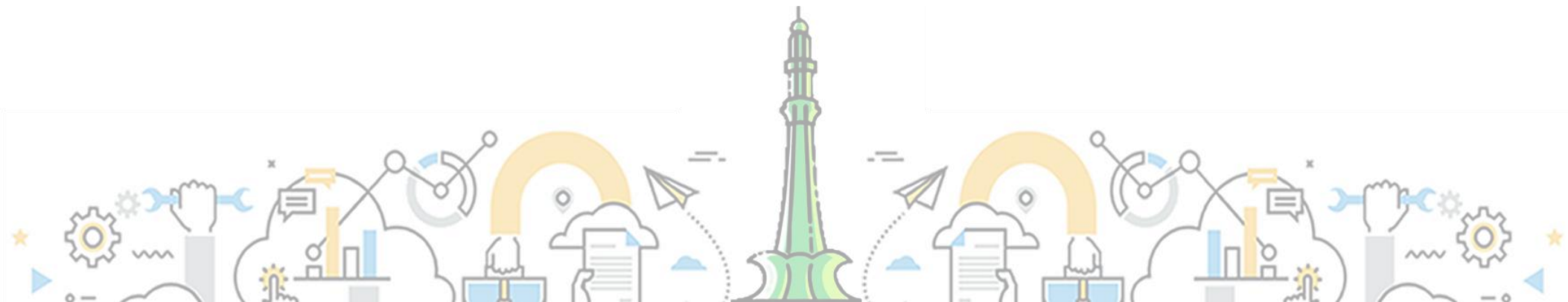




Requirement Pre-Requisites

Solution 4/4

4. THINKING THAT AGILE IS ABOUT STICKING TO THE MANIFESTO AND THERE IS NO ROOM FOR INOVATION, ADAPTABILITY OR CHANGE.





Requirement Pre-Requisites

SUMMARY FOR REQUIREMENT GATHERING

1. We need to Document the requirement to ensure an intended product is built. How you document depends on What suits you and your Project.
2. Communication Skills is very important. Just Speaking the Language does not qualify that Communication is being done. It is a multidirectional skill





Requirement Pre-Requisites

SUMMARY FOR REQUIREMENT GATHERING (Cont.)

3. Be a part of the Requirements. Question the Requirements and the about the requirements.
4. Agile is a mind set and is meant to be fluid. Mold it to what your project needs and what the environment is and not the other way around.





Requirement Pre-Requisites

The Obvious answer is wrong

